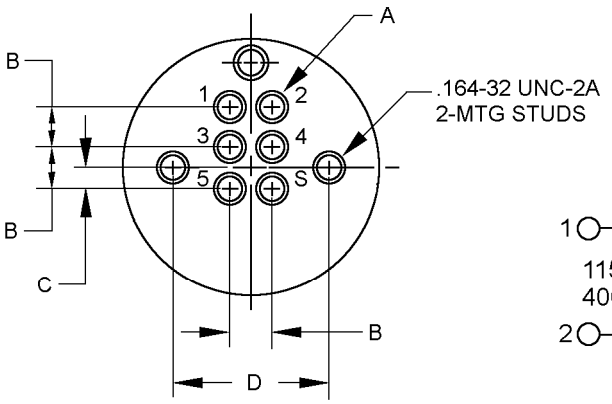
INCH-POUND
MIL-PRF-27/61D
AMENDMENT 1
12 August 2016
SUPERSEDING
MIL-PRF-27/61D
6 May 2005

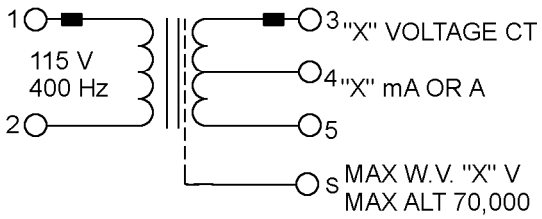
PERFORMANCE SPECIFICATION SHEET

TRANSFORMERS, POWER, 16 VOLT AMPERES, 400 HZ, ELECTROSTATIC SHIELDED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

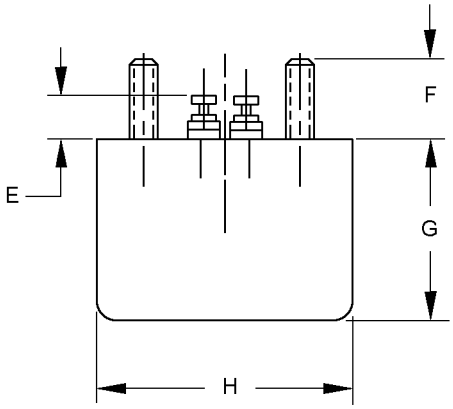
The requirements for acquiring the transformers described herein shall consist of this specification and MIL-PRF-27.





(SEE TABLE I FOR VALUE OF "X")

CIRCUIT DIAGRAM AND MARKING



Dim	Inches	mm
A	.234 DIA. MAX.	5.94 DIA. MAX.
B	.250	6.35
C	.125	3.18
D	1.000 ±.031	25.40 ±.078
E	.312 MAX.	7.92 MAX.
F	.344 ±.031	8.74 ±.078
G	1.125 ±.031	28.58 ±.078

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Tolerance on terminal position dimensions is ±.015 (0.38 mm).
4. Marking shall be on the side of the case.
5. Terminals are color coded for permanent identification in addition to terminal marking shown. Ground terminal "S" is black.
6. Secondary electrical values shall be marked as specified in table I, as applicable.

FIGURE 1. Dimensions and configuration.

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REQUIREMENTS: (When numbers in parentheses i.e. (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings: See table I.

Primary voltage (1-2): 115 volts ±10 percent, 400 Hz ±5 percent.

Design and construction:

Dimensions and configuration: See figure 1.

Weight: 3.8 ounces, maximum.

Duty cycle: Continuous.

Case: Metal enclosed.

Material: Steel.

Altitude: 70,000 feet, maximum.

Operating temperature range: -55°C to +130°C maximum.

Terminals: Feed thru.

Terminal height: .312 inch (7.92 mm), maximum.

Terminal strength: MIL-STD-202-211, test condition A, 5 pounds.

Dielectric withstanding voltage:

At sea level: 2.8 times the maximum working voltage specified in table I.

At reduced barometric pressure: 1.25 times the maximum working voltage specified in table I.

Electrical characteristics:

No load: With 115 volts, 400 Hz across (1-2): Current in (1-2) shall not exceed 16 mA rms.

Rated load: With 115 volts, 400 Hz across (1-2) and a resistive load with current as specified in table I in (3-5), the voltage across (3-5) shall be as specified in table I.

Efficiency: 87 percent minimum at rated load.

Electrostatic shielding: The voltage ratio shall be a minimum of 5 to 1 at 20 kHz.

Polarity: Additive with terminals 2 and 3 connected.

Regulation: $\frac{Voltage\ (no\ load) - Voltage\ (rated\ load)}{Voltage\ (rated\ load)} \quad x \quad 100$

Shall not exceed 8 percent.

Phase shift: 1.0 electrical degree between primary and secondary voltage at rated load, maximum.

Temperature rise: 32°C, maximum with 115 volts, 400 Hz across (1-2) at ambient temperature of 98°C.

Vibration: MIL-STD-202-204, test condition D.

Marking location: See figure 1.

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VERIFICATION:

Extent of qualification:

Qualification testing and approval to M27/49-44 A or B, M27/51-44 A or B, M27/53-44 A or B, M27/56-44 and M27/66-27 shall be sufficient to grant qualification approval to MIL-PRF-27/48 through MIL-PRF-27/66 inclusive, all parts.

Qualification testing and approval to M27/61-44 shall be sufficient to grant qualification approval to M27/61-01 through M27/61-44.

Part or Identifying Number (PIN) example:

M	27/61	-01
MIL specification item	Specification sheet no.	Dash no.

TABLE I. Electrical ratings.

Dash number	Secondary voltage (3-5) (V rms) ±3%	Secondary current (3-5) (amps rms)	Working voltage (V rms max)
01	6	2.667	325
02	6.3	2.540	325
03	7	2.286	325
04	7.5	2.133	325
05	8	2.000	325
06	9	1.778	325
07	10	1.600	325
08	11	1.455	325
09	12	1.333	325
10	12.6	1.270	325
11	13	1.231	325
12	15	1.067	325
13	16	1.00	325
14	18	0.889	325
15	20	0.800	325
16	22	0.727	325
17	24	0.667	325
18	26	0.615	325
19	28	0.571	325
20	30	0.533	325
21	33	0.485	325
22	36	0.444	325
23	39	0.410	325
24	43	0.372	325
25	47	0.340	325
26	51	0.314	325
27	56	0.286	325
28	62	0.258	325
29	68	0.236	325
30	75	0.213	325
31	82	0.195	325
32	91	0.176	325
33	100	0.160	325
34	110	0.145	325
35	115	0.139	325
36	120	0.133	325
37	130	0.123	325
38	150	0.106	325
39	180	0.089	325
40	200	0.080	440
41	220	0.073	440
42	240	0.067	440
43	270	0.059	580
44	300	0.053	580

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Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202-204, MIL-STD-202-211

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5950-2016-011)

Review activities:
Army – CR4, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at https://assist.dla.mil.

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